

## QUESTION OF THE WEEK 5

### GRADE XI CHEMISTRY

#### Assignment Submission Date: Bhadra 1

1. Differentiate between electron affinity and electronegativity.
2. Compare the size of  $S^{2-}$ ,  $Cl^-$ ,  $Ar$ ,  $K^+$  and  $Ca^{++}$  with reason.
3. Define electron affinity. What are the various factors that affect electron affinity?
4. State and explain Graham's Law of Diffusion. If the time of equal effusion of pure oxygen and a mixture of oxygen and ozone are 224 sec and 243 sec respectively, find the percentage of ozone in the mixture.
5. State and explain Dalton's Law of partial pressure. 500 cc of oxygen under 680 mm pressure and 250 cc of nitrogen at 720 mm pressure were put together in a litre flask keeping the temperature constant. Find the final pressure of the mixture.
6. Balance the following redox reaction by ion-electron method.
  - a)  $Cl_2 + OH^- \longrightarrow Cl^- + ClO_3^- + H_2O$
  - b)  $MnO_4^- + H_2O_2 + H^+ \longrightarrow MnO_2 + O_2 + H_2O$